

Shengda LoRaWAN

HAC-MLW Application Layer Protocol V2.1

1. Protocol format

There can be one or more sets of TV format data items in one data packet.

Frame header	T	V	CS
(1 byte)	Data length flag	Data type	1 byte

Remark: the value of V FF is the reading, e g: 0B FF FF FF FF is to read the pulse number, 06 FF FF FF is to read the temperature.

2. Detailed explanation of TV data

	Data flag (T)	Data type (V)	Length	Read&write	fixed unit
	0x01	Current cooling amount (the heating on settlement day)	4	R	kWh
	0x02	Current heat amount	4	R	kWh
	0x03	Heat power	4	R	kW
	0x04	Instantaneous flow	4	R	L/h
	0x05	Data length	1	R	
	0x06	Supplying water temperature	3	R	0.01℃
	0x07	Returning water temperature	3	R	0.01℃
	0x08	Cumulative working time	3	R	h
	0x09	LORAWAN working mode	1	RW	
	0x0A	Sensor voltage	2	R	mv
	0x0B	Number of pulse	4	RW	pulse
	0x0C	Daily accumulated flow	4	R	pulse
	0x0D	Monthly frozen cumulative flow	4	R	pulse
	0x0E	Yearly frozen cumulative flow	4	R	pulse
	0x0F	Settlement date	1	RW	None
	0x10	Reset	1	RW	None
	0x11	LoRaWAN ClassB downlink communication cycle	1	RW	None
	0x12	Metering mode	1	RW	None
	0x13	Maximum metering value	4	RW	None
	0x14	Pulse constant	1	RW	None
	0x15	RSSI value	1	R	DBm
	0x16	Device (S/N)	4	RW	None

	0x17	Valve type	1	RW	None
	0x18	Device version information	n	R	None
	0x19	Packet sequence	1	R	None
	0x1A	Battery voltage	2	R	mV
	0x1B	Meter type	1	RW	None
	0x1C	Module time	6	RW	None
	0x1D	Frozen data time (year, month, day)	3	R	None
	0x1E	Device address (Device EUI)	8	RW	None
	0x1F	Valve control	1	W	None
	0x20	Remote reading	1	R	None
	0x21	Current cumulative reverse flow	4	R	L
	0x22	Historical flow record	n	R	
	0x23	Trigger source	1	R	None
	0x24	Maximum valve control time	1	RW	S
	0x25	Data report interval	4	RW	S
	0x26	Overcurrent value	4	RW	L/H
	0x27	Duration of overcurrent	4	RW	H
	0x28	Continuous time of water usage	4	RW	H
	0x29	The maximum amount of	4	RW	L
		water usage for a single time			
	0x2A	The maximum value of continuous reverse	4	RW	L
	0x2B	The starting time of reporting	1	RW	H
	0x2C	Dense sampling period	2	RW	Min (the specified value)
	0x2D	Q3 value of water meter	2	RW	L/H
	0x2E	Enable reporting LoRaWAN	1	RW	0 by default (H73B only)
	0x2F	Read the 3-way maximum value of non-magnetic inductive	1/8	R	MLWA project only
	0x30	Minimum flow	2	R	L/H
	0x31	Maximum flow	2	R	L/H
	0x32	NULL			F14 occupied
	0x33	Status word	2	R	None
	0x34	Device (APPEUI)	8	RW	None
	0x35	Device (APPKEY)	16	RW	None
	0x36	Center frequency offset	2	RW	Hz
	0x37	Switch on/off	1	RW	None
	0x38	Total power	4	R	kwh
	0x39	A phase current	2		
	0x3A	B phase current	2		
	0x3B	C phase current	2		

	0x3C	A phase active power	2		
	0x3D	B phase active power	2		
	0x3E	C phase active power	2		
	0x3F	Channel information	1	RW	
	0x40	Pressure value	2	RW	Kpa
	0x41	Pulse can be used	4	RW	
	0x42	Switch open or close setting for threshold	5	RW	
	0x43	Frozen data time(year month day hour minute)	5/9	RW	Dense frozen is used
	0x44	Alarm report period	4	RW	s
	0x45	Transmission power setting	1	RW	dbm
	0x46	DEV_ADDR	4	RW	
	0x47	NWK_SKEY	16	RW	
	0x48	APP_SKEY	16	RW	
	0x49	Instantaneous velocity	4	R	m/s
	0x51	Time zone	1	RW	One byte have symbol type
	0x52	Lifespan for using	2	RW	
	0x53	Downlink delay time	1	RW	s
	0x54	Number of reports	4	RW	
Special using	0x90	Transparent transmission	n	RW	2490020101B8 02:L Total length of transparent transmission 0101: transparent transmission data
	0x5048	Upgrade command	5	W	504841434D4C57

Remark:

1. All data transmission adopts HEX code except for reading version information,

2. All data are in Big-Endian mode.

3. LoRaWAN working mode (T = 0X09)

- a) 0X00: CLASS A
- b) 0X01: CLASS B
- c) 0X02: CLASS C
- d) 0X03: LoRaWAN dual communication mode

4. LoRaWAN ClassB downlink communication cycle (T = 0x11)

WR can be set in the range of 0~7, integer type, which is represented by X below:

The actual period of LoRaWAN ClassB downlink communication = $122880 / (128 / (2^X))$

ms allows to send data once

5. Reset (T = 0x10)

- a) 0x00~0xfe: number of reset
- a) 0xff: system reset(the module will make system reset automatically once receiving the command)

6. Metering mode: (T = 0X12)

- a) 0x00: dual reed switch;
- b) 0x01: single reed switch;

- c)0x02: dual hall;
- d)0x03: meter reading directly
- e)0x04: non-magnetic inductive
- f)0x05: non-magnetic coil
- g)0x06: three hall
- h)0x07: single hall
- i)0x08: EDC-U pulse
- j)0x09: IUW pulse
- k)0x0A: EDC-B1 pulse
- l)0x0B: EDC-B2 pulse
- m)0x0C: IUW-NFC pulse
- n)0x0D: ADC acquisition
- o)0x0e: Near camera direct reading
- p)0x0f: Remote camera direct reading
- q)0x10~0xff: Reserve

7. Pulse constant (T = 0X14)

- a) PN=0x00: 1 metering pulse is 0.5 L;
- b) PN=0x01: 1 metering pulse is 1 L;
- c) PN=0x02: 1 metering pulse is 10L;
- d) PN=0x03: 1 metering pulse is 100 L;
- e) PN=0x04: 1 metering pulse is 1000L;
- f) PN=0x05: 1 metering pulse is 10000L;
- g) PN=0x06: 1 metering pulse is 5L;
- h) PN=0x07~0xff: Reserve

8. Valve type (T = 0x17)

- a)0x00: Two-wire valve
- b)0x01: Five-wire valve
- c)0x02: No valve
- d)0x03: Angle valve
- e)0x04: Four-wire valve

9. Meter type(T = 0X1B)

- a) 0x00: Water meter
- b) 0x01: Gas meter
- c) 0x02: Heat meter
- d) 0x03: Electricity meter
- e) 0x04: Gas sensor
- f) 0x05: smoke detector
- g) 0x06: electromagnetic flow meter
- h) 0x07-0xFE: Reserve

10. Trigger source (T = 0X23)

- a) 0x00 : Magnetic trigger reporting
- b) 0x01: Routine reporting
- c) 0x02: Magnetic attack reporting
- d) 0x03: Valve control reporting
- e) 0x04: The platform reads the sensor information and reports it
- f) 0x05: The platform reads and reports the version information
- g) 0x06: The platform sets the parameter and reports
- h) 0x07: Monthly frozen
- i)0x08: Yearly frozen
- j)0x09: Network successfully reported
- k)0x0A: Dredge valve report
- l)0x0B: Modification reporting of network join parameters

- m)0x0C: Modification reporting of parameters such as valve type and frequency offset
- n)0x0D: Upgrade command reporting
- o)0x0E: Reporting interval reporting
- p)0x0F: Non-magnetic inductive alarm parameters reporting
- q)0x10: Dense sampling cycle reporting
- r)0x11: Q3 valve reporting
- s)0x12: Start LoRaWAN to report
- t)0x13: Abnormal alarm reporting
- u)0x14: LoRaWAN working mode modification reporting
- v)0x15: Sensor status feedback package
- w)0x16: Touch button trigger reporting
- x)0x17: Threshold parameter modification reporting
- y)0xFF: Parameter error package

If use the hall sensor, the trigger source is 0x00, if use Touch button, the trigger source is 0x16

11. Valve control (T = 0X1F)

- a)0X00: Open the valve
- b)0x01: Close the valve
- c)0x02: Dredge the valve
- d)0x03: The regular dredging valve function is turned on
- e)0x04: The regular dredging valve function is turned off
- f)0x05: The function of valve closed automatically if cut off power is turned on
- g)0x06: The function of valve closed automatically if cut off power is turned off
- h)0x64(100)~0xc8(200): angle valve control, 0x64: open 0% , 0xc8: open 100%

12. Remote meter reading (T = 0X20)

- a)0X01: Meter information reading

13. Switch on/off (T = 0x37)

- a)0x01 Switch on
- b)0x02 Switch off

14. Battery voltage

Two-byte means, such as 241A003c; data 0x003c, which is converted to decimal 60, that is, battery voltage is 60/16.4V=3.66V.

15. System time

Format: year, month, day, hour, minute, second, each one byte, the year is based on 2000

For example: 241C130301000000 is March 1, 2019 00:00:00

24: frame header

1C: data type

13: which is corresponding to decimal 19, which is 2019

03: 3 month

00: 0 hour

00: 0 minute

00: 0 second

16. CS check

CS is cumulative sum check, the cumulative sum of all other bytes except CS check.

17. Regularly report data packets

Uploaded data type packet sequence, meter S/N, meter type, metering mode, metering value(number of pulse), pulse constant, battery voltage, status word, trigger source.

18. Reporting interval (data 0x25)

WR it can be set in the range 600 ~86400, integer, unit: s(second).

19. The reporting starting time (data 0x2B)

WR it can be set in the range 0~23, integer, unit: h

20. Dense sampling period (data 0x2C)

WR it can be set as a submultiple of 60 or a multiple of 60, 1-0x90 which is 5, 10, 15, 20, 30, 60, 120, 240, 360, 720, 1440 min

21. Q3 value (data 0x2D)

WR it can be set in the range of 0~65534, the unit is L/H.

22. Start LoRaWAN reporting (data 0x2E)

W Data item 1 byte (reserved) Fixed 00, which is only applicable to infrared long-timeworking module (H73B)

23. Channel information (0x3F)

(Bit4~Bit7) working frequency	(Bit0~Bit3) channel group
0:EU433 1:CN470 2:IN868 3:EU868 4:AU915 5:US915 6:AS923 7~15:reserve	

24. Historical flow records (data ID 0xA2)

Storage period 1~0x90, the actual value is 5 times of the reported value. Reported value: 0x01, which means 1*5min, 0x90 means 0x90*5 = 0x2d0min (12h)

Storage period 0x90~0xff, the actual value is 10 times of the reported value. Reported value: 0xd8, which means (0x90*5)+(0xd8-0x90)*0x0a=0x5a0(24h)

The specific TLV format of historical flow records are as below:

	T	L	V						
Expla nation	Attribute	Data length	1 byte (storage cycle)	4 -byte Storage initial water consumption	2- byte N1	2-byte N2	2-byte N3	2-byte N4	2-byte ...

N1, N2, N3, N4...: means the measurement difference valve between the next and the previous one.

The highest bit of N1, N2, N3, and N4 is 1, indicating that the difference is negative (The latest water consumption is less than the last storage water consumption).

Eg: 0xA2 0x0F 0x01 0x00 0x00 0x00 0x32 0x00 0x01 0x00 0x02 0x80 0x03 0x00 0x04 0x80 0x05

0xA2	0x0F	0x01	0x00 0x00 0x00 0x32	0x00 0x01	0x00 0x02	0x80 0x03	0x00 0x04	0x80 0x05
T	L	5 Min to store data once	Storage starting metering value 0x32liters	After 5 min of Storage starting 0x33liters	After 10 min of Storage starting 0x35liters	After 15 min of Storage starting 0x32liter s	After 20 min of Storage starting 0x36 liters	After 25 min of Storage starting 0x31 liters

3. Meter Status Word

1. Water meter

Meter status word 1

	B7	B6	B5	B4	B3	B2	B1	B0
Definition	Valve fault	Battery status	Magnetic attack/temperature alarm	Battery power off	DER	Valve status	Metering failure (remove off)/EE fault	Reserve
Explanation	0: Normal 1: Fault	0: Normal 1: lack voltage	0: Normal 1: Fault	0: Normal 1: The battery has been removed off.	0: Correct 1: Wrong	0: Open 1: Close	0: Normal (not disassembled) 1: Fault (disassembled)	

Among:

B₇ Fault flag: 0: normal; 1: valve fault, including in-position switch fault, low voltage when controlling the valve or the valve itself does not turn.

B₆ Battery voltage: 0: normal; 1: lack voltage (meter battery voltage status).

B₅ Magnetic attack flag/temperature alarm: 0: normal; 1: magnetic attack.

B₄ Power cut off flag: 0: The battery is normal; 1: The battery has been removed.

B₃ DER status: 0: The current measurement data is correct; 1: The current measurement data is wrong.

B₂ Valve status: 0: valve open; 1: valve close

B₁ Metering fault flag/EE fault: 0: normal; 1: metering open circuit fault
(J51I means the dismantling position)

B₀ Remote flag: 0: the data of module; 1: remote data

Meter Status Word 2

	B7	B6	B5	B4	B3	B2	B1	B0
Definition	Water inlet alarm	Water return alarm	Flow alarm (historical dismantling)/over range alarm	Meter status	leakage	Tube burst	Historical magnetic attack/reverse flow alarm	Remote flag
Explanation	0: normal 1: fault	0: normal 1: fault	0: normal (no history dismantle) 1: fault (historical dismantle)	00: Normal 01: Empty pipe	0: normal 1: fault	0: normal 1: fault	0: normal 1: there is magnetic attack	0: module data 1: remote data

2. Gas sensor

Meter status word 1

	B7	B6	B5	B4	B3	B2	B1	B0
Definition	Preheat	Battery status	Sensor life	Battery power off	DER	Valve status	Metering failure (remove off)	Remote flag

Explanation	0:Normal 1:Fault	0:Normal 1:Under voltage	0:Normal 1:End	0:Normal 1:The battery has been removed off.	0:Correct 1:Wrong	0:Open 1:Close	0:Normal (not disassembled) 1:Fault (disassembly)	0:Moduledata 1:Remotedata
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Meter Status Word 2

	B7	B6	B5	B4	B3	B2	B1	B0
Definition	Concentrationalarm	Fault alarm	Reserve					
Explanation	0:normal 1:fault	0:normal 1:fault	Reserve					

4. Conditions for timing calibrating transmitted by the platform

The module reports regularly every day. In order to ensure that the time deviation is small, the platform needs to transmit a time checking command to calibrate the module. The conditions for the platform to transmit the time calibrating are:

1. Complete uplink data and the decode is normal.

The trigger source of data analysis is the magnetic trigger report, the routine report or the dredge valve report to transmit the time calibrating command, and other data will not transmit the command of time calibrating.

5. Example of transmitting the data

Note: The last digit of the data in the example is the CS verification data, it is needed when command with CS verification.

1. Monthly frozen cumulative flow

260D13044A (read monthly frozen data in April 2019)

Frame header + data type + year + month

Return the data: 24190d1d1304000d0000c35014032307df

Data analysis: the 13th packet of data, the frozen data time is April 2019, the monthly frozen flow is 5000, 100L/P, the monthly frozen data

2. Annual frozen cumulative flow

260E1347 (read 2019 frozen data)

Frame header + data type + year

Return the data: 2419081d1300000e0001d4c0140323085a

Data analysis: the 8th packet of data, the frozen data time is 2019, the annual frozen flow is 120,000, 100L/P, and the annual frozen data

Note: If the frozen data time is 0x00, there is no need to decode.

3. Version Information

26183E (read module version information), the version information uploaded is ASCII code

The data after reading the version information is:

Return the data:

Data analysis: the second packet data, HAC-MLW-C V1.0.0, version information

4. Read module time

261C42 (read module time)

Return the data: 2419041c1301010101052304a0

Data analysis: the 4th packet data, 01:01:05 on January 1, 2019

5. Write module time

261C130101010105FF (Write module time)

Return the data: 2419041c1301010101052304a0

Data analysis: The 4th packet of data, at 01:01:05 on January 1, 2019, read sensor information

6. Report the calibrating time of module

261C130101010105FF (Writing the time of module: January 1, 2019, 01:01:01)

Frame header + data type + year + month + day + hour + minute + second + CS

Year calculation: 0x13 which corresponds to decimal 19, year = 2000 + 19 = 2019, the year of the data is the current year minus the current year value of 2000

7. Valve control

Open the valve: 261F0045

Close the valve: 261F0146

Dredge the valve: 261F0247

Regular dredge valve function is turn on: 261F0348

Regular dredge valve function is turn off: 261F0449

8. Parameter setting

Set the metering value:

26130001869f14030B00000000FF (the maximum metering value is 9999900, the pulse constant is 100L/P, the metering value is 0)

Set meter type:

261B0142 (gas meter)

Set the metering mode:

2612023A (hall metering)

Set the report interval:

262500000E1069 (the reporting interval is 1H)

Historical flow record reporting: (12 frozen data reporting)

24 1605f5e104 1401 0b00018623 A21B0C 00000000 0001 0002 0003 0004 0005 0006 0007 0008 0009 000A 000B 1a003c 330000 2301 CS

Set the Q3 value

261E81234500000000012D00015C

2419021E81234500000000012D0001231115D08E

Start LoRaWAN reporting

261E81234500000000012E005C

2419021E81234500000000012E00231215D08F

6. Special usage

Transparent transmission command: (TLV)

26 90 12 FEF68AAAAAAAAAAAAAAAAA0303810A004916 6C

(Transparent transmission content FEF68AAAAAAAAAAAAAAAAA0303810A004916)

Upgrade command: (T(T0T1)V)

26504841434D4C5732

Note: The frame header is 0x26, which is distinguished from the frameheader of the reporting.